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SIVAGANGAI

S **COMMON HALF YEARLY EXAMINATION - 2025**

Standard - 12 Reg.No. | | | | | | |

Time Allowed:3.00 Hours **BIOLOGY** *Maximum Marks:70*

BIO-BOTANY

Part-I

I. Answer the following questions: **8×1=8**

1) Which of the following explain how progeny can posses the combination of trait that none of the parent possed?
   a) Law of segregation b) Chromosome theory
   c) Law of independent assortment d) Polygenic inheritance
2) The prevention of large scale loss of biological interity
   a) Biopatent b) Bioethics c) Biosafety d) Biofuel
3) The plant known as the Terror of Bengal
   a) Pistia b) Chrysanthemum
   c) Dioscorea d) Eichhorniacrassipes
4) Solar energy used by green plants for photosynthesis is only
   a) 2-8% b) 2-10% c) 3-10% d) 2-9%
5) Cross between the plants of the same variety are called
   a) interspecific b) inter varietal c) intra varietal d) inter generic
6) Tectonagrandisis coming under family
   a) Lamiaceae b) Fabaceae
   c) Dipterocaipaceae d) Ebenaceae
7) _____ mediated genetic engineering technique was followed to produce Flavr - Savr tomato.
   a) Bacillus thuringiensis b) Agrobacterium
   c) Lepidopteran d) all the above
8) The indication of Sulphur di oxide is
   a) Lichens b) Petunia c) Chrysanthemum d) Gladiolus

Part-II

II. Answer any four questions: **4×2=8**

9) What are the materials used to grow microorganism like spirulina?
10) What is bio fertilizers?
11) What is photosynthetically Active Radiation?
12) Draw a structure of embryo sac and label the parts.
13) Differentiate bio medicines and botanical medicines.
14) Write the various steps involved in cell suspension culture?

Part-III

III. Answer any three questions: (Q.No:19 is compulsory) **3×3=9**

15) What is reciprocal cross?
16) What is gene mapping? Write its uses.
17) Distinguish habitat and niche.
18) What is CCS?
19) Enumerate the uses of turmeric?

Part-IV

IV. Answer the following questions: **2×5=10**

20) Give a detailed account on parthenocarpy. Add a note on its significance.

(OR)

What are the steps involved in Protoplast culture?

21) How do scared groves help in the conservation of biodiversity?

(OR)

What is soil profile? Explain the characters of different soil horizones?

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BIO-ZOOLOGY

I. Choose the correct answer: 8×1=8

1) In which type of parthenogenesis males and females are produced?
   a) Thelytoky b) Arrhenotoky
   c) Amphitoky d) Both (a) and (b)
2) The process which the sperm undergoes before penetrating the ovum is
   a) Spermiation b) Capacitation
   c) Cortical reaction d) Spermiogenesis
3) ZO-type and ZW-type of Sex determination are examples of
   a) Male heterogamety b) Female heterogamety
   c) Female homogamety d) Both (a) and (b)
4) Modern man belongs to which period?
   a) Quaternary b) Cretaceous c) Silurian d) Cambrian
5) Parotop is an
   a) Antibody binding site on variable regions
   b) Antibody binding site on heavy regions
   c) Antigen binding site on variable regions
   d) Antigen binding site on heavy regions
6) Some organisms are able to maintain homeostasis by physical means
   a) Conform b) Regulate c) Migrate d) Suspend
7) Which one of the following is not comming under insitu conservation?
   a) Sanctuaries b) Natural parks
   c) Zoological park d) Biosphere reserve
8) In the E-Waste generated by the Mobile Phones, which among the following metal is most abundunt?
   a) Copper b) Silver c) Palladium d) Gold

II. Answer any four of the following questions: 4×2=8

9) Why is the offspring formed by asexual reproduction referred as a clone?
10) What are the applecations of Karyotyping?
11) Why the human genome project is called a mega project?
12) Define the following terms: a) Zymology b) Superbug
13) What are called endangered species?
14) What are DNA vaccines?

III. Answer any three questions: (Q.No.19 is compulsory) 3×3=9

15) Write any three Assisted Reproductive Technology.
16) Explain any three microorganisms involved in bio-remediation.
17) Differentiate between somatic cell gene therapy and germline gene therapy.
18) Write the various differences between r-selected and k-selected species.
19) Write about the three indices of species diversity.

IV. Answer all the questions: 2×5=10

20) a) Explain the various phases of the menstrual cycle.

(OR)

b) Explain the classical model of Lac operon proposed by Jacob and Monod.

21) a) Tabulate the types of Innate immunity and its mechanisms.

(OR)

b) Explain the process of sewage treatment.

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